AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I046*

Team Name: *Closed Loop*

Domain: *8*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Impacts and Benefits
11. Sustainable Development Goals
12. Individual Member Contribution

Problem Statement

Commercial logistics operations encounter severe bottlenecks in last-mile delivery. Ground vehicles experience urban congestion. This increases delivery times, fuel costs, and carbon emissions.

Current UAV delivery solutions have strict physical limitations:
- **Fixed-wing UAVs:** Provide long range and high speed, but cannot navigate complex urban environments or perform precise package drops.
- **Multirotor UAVs:** Provide precision hover capability, but operate with severe limitations in battery life, range, and payload capacity. They also create high acoustic pollution.

Problem Understanding

- **Who experiences this problem?**
  - Commercial Logistics Providers (3PLs)
  - Major E-Commerce Companies (Amazon, Flipkart)

- **What causes the problem?**
  - Severe urban traffic congestion and limited ground parking.
  - Regulatory bottlenecks in aerial drone delivery (Noise limits).
  - Extremely high power-draw for drones carrying heavy cargo over long distances.

- **How does it affect users or stakeholders?**
  - Last-mile delivery accounts for up to 53% of total shipping costs.
  - High rates of Return-to-Origin (RTO) due to missed delivery windows.
  - E-commerce brands suffer huge losses in urban real estate for micro-warehouses.

Existing Gaps & Challenges

- **Existing Limitation 1: Point-to-Point Inefficiency**
  - Current delivery drones (e.g., Wing) carry only 1–2 small parcels per 20km flight. They lack the payload capacity to scale economically for bulk e-commerce.

- **Existing Limitation 2: Ground Traffic Gridlock**
  - Traditional delivery vans average under 15 km/h in Tier-1 cities, creating massive delays and high carbon emissions.

- **Main Challenge that Remains Unsolved**
  - Every existing aerial logistics solution can deliver single package in a single mission. No single Aerial Logistics system supports multi package delivery.

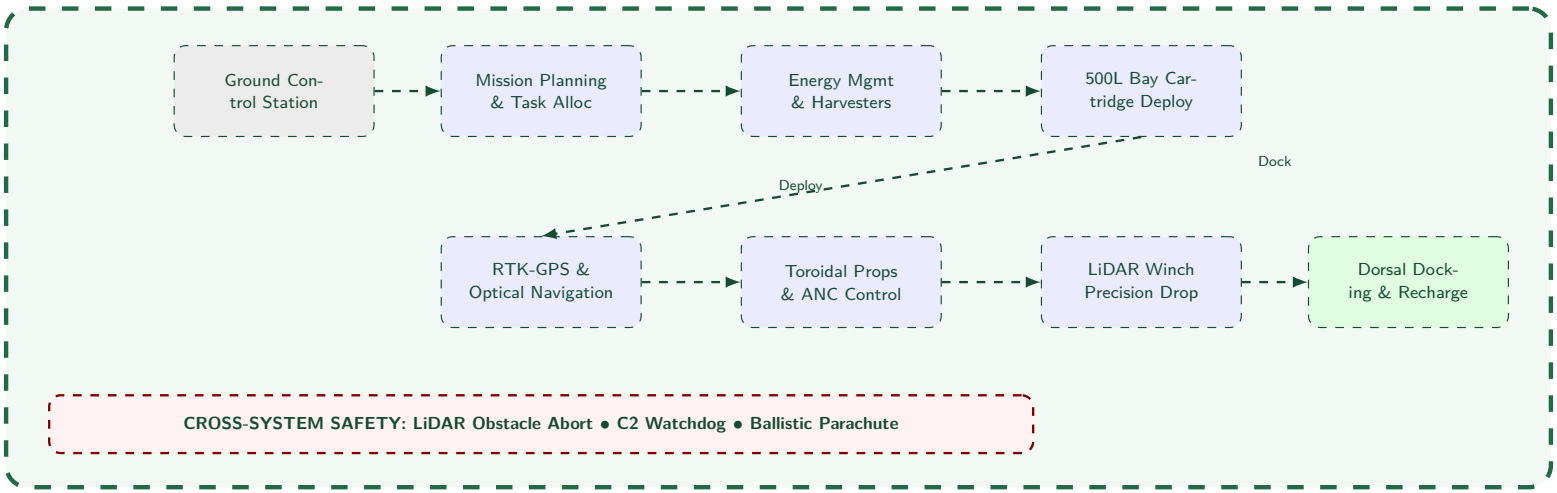
Our Proposed Solution

- **Core Idea: Project SkyHive**
  - An Autonomous two-tiered UAV network that completely separates the long-distance transportation phase from the localized urban delivery phase.

- **How the solution works at a high level**
  - *Hub Fixed-Wing VTOL (Mothership):* Acts as a mobile, airborne distribution hub holding up to 200kg of payload, hovering safely above the city.
  - *Micro-Swarm Deployment:* The Mothership simultaneously deploys 16 multirotor drones to perform precise, localized package drops.
  - *Automated Mid-Air Recovery:* Drones ascend, dock onto the Mothership, and trickle-charge in flight before the next sector.

- **Main Value Delivered to Users**
  - Slashes last-mile costs, bypasses ground traffic, and acts as a zero-rent mobile micro-fulfillment center for logistics companies.

Technical Approach



The Mothership acts as a high-altitude C2 relay. It deploys the micro-swarm via a cartridge system. The swarm uses GPS and LiDAR for precise drops, then returns for automated dorsal docking.

Tech Stack

Frontend: QGroundControl (GCS UI)

Backend: ROS2 Humble

Database: PostgreSQL (Fleet Logs)

AI,ML,API: OpenCV, PX4, MAVSDK

Innovation & Uniqueness

- **First-of-its-Kind Multi-Package Sortie:** Current aerial logistics are heavily constrained by a 1-to-1 ratio (one drone delivers one package per trip). SkyHive shatters this limitation by successfully delivering 16 distinct packages in a single, continuous mission.

- **Mobile Airborne Micro-Fulfillment:** We eliminate the need for individual drones to fly back and forth from a distant warehouse. The Mothership acts as a flying hub, transporting the entire payload inventory directly over the target urban sector.

- **Parallel Swarm Deployment:** Unlike traditional delivery vans that must make sequential stops, or single-payload drones that waste battery on long transits, our micro-swarm deploys simultaneously to drop multiple packages across a 10 km radius and re-docks mid-air.

Feasibility & Scalability

- **Feasibility & Architecture Alignment:** The 11% payload mass fraction (200kg load / 1800kg MTOW) aligns with established Lift-plus-Cruise eVTOL aerospace benchmarks. We utilize mature COTS avionics (Pixhawk), open-source middleware (ROS2/PX4).

- **Scalability (Logistics-as-a-Service):** Highly scalable via a B2B model. A single hub covers a 100km radius. Swarm configurations (mixing 2.5kg and heavy-lift 7.5kg payload drones) can be dynamically adjusted per Mothership based on daily urban commercial demand.

- **Risks & Mitigations:**
  - *Risk:* Mid-Air Rendezvous Docking Collision.
  - *Mitigation:* Implement a multi-stage recovery pipeline. For terminal capture, use downward-facing optical cameras tracking pulsed IR LED fiducials and ArUco markers on the Mothership deck, finalized by magnetic self-centering rings to physically lock the drones in place.
  - *Risk:* Dynamic ground obstacles.
  - *Mitigation:* Downward single-beam LiDAR triggers a smart-abort sequence (30m climb) if a human or vehicle enters the drop zone.

Cost for Product

Approx. Projections-

- **R&D and Prototype Engineering (NRE):** 1.5 Crores
  - Includes ROS2 Swarm logic, CFD analysis, wind-tunnel testing, and custom carbon-fiber layup tooling for the Mothership.
- **Scaled Hardware Production (COGS):** 2.5 Crores per unit
  - Manufacturing one fully integrated system at volume (1 Mothership + 12 Micro-Drones).
- **Commercial Strategy (Logistics-as-a-Service):**
  - Instead of selling the aircraft directly, we operate the fleet and charge e-commerce partners a flat rate of **300rs to 400rs per successful drop**.

Use Cases

- **Use Case 1: E-Commerce Same-Day Urban Delivery**
  Amazon or Flipkart routes 16 standard parcels (up to 2.5kg each) to the Mothership. The system bypasses rush-hour traffic to complete all residential doorstep drops in 20 minutes, vastly reducing delivery delays.

- **Use Case 2: Food Delivery/Porter Services**
  Instead of delivering 1 or 2 food packages at a time we can circulate almost 16 packages of food for the given area which reduces the required man power as well as reduces the usage of urban road implying reduction in congestion

- **Use Case 3: Emergency Medical & Blood Supply**
  During a crisis, the Mothership loads cold-chain blood packs and deploys the swarm in parallel, delivering critical supplies to multiple localized clinics simultaneously across a 10km disaster radius.

Impacts and Benefits

- **Benefit to Logistics Operators:**
  Slashes last-mile overhead costs and virtually eliminates the Return-to-Origin (RTO) loss by guaranteeing delivery windows regardless of ground gridlock.

- **Social & Environmental Impact:**
  Replaces thousands of diesel delivery vans on urban streets. The completely electric powertrain and solar PV harvesting dramatically lower the carbon footprint of commercial retail.

- **Expected Measurable Outcome:**
  A single Mothership completing 10 daily sorties can successfully process 100+ packages, effectively replacing around 10 to 15 traditional delivery vehicles per unit.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 7	Target 7.2	Affordable & Clean Energy	Solar PV arrays and wing-tip turbines.
SDG 9	Target 9.4	Industry, Innovation & Infrastructure	Complete Autonomous Aerial Logistics Network
SDG 11	Target 11.6	Sustainable Cities	Reduces urban ground traffic congestion.
SDG 13	Target 13.2	Climate Action	Electric systems & solar harvesting lower carbon footprint.

Individual Member Contribution

Member	Contribution or responsibility
K Lokesh	Team Lead & Flight Ops:System integration, resource allocation, and flight test safety protocols.
Harish B	Software Lead: Autonomous swarm flight control, computer vision, and GPS, Navigation integration.
GuruSharvesh B	Hardware Lead: CAD airframe design, propulsion, fabrication, and energy harvesting integration.
Rahul Kumar	Technical Lead: Systems architecture, SITL simulation pipelines, and C2 relay networking.

Yukti ID

IR2026-1051027